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Hypothetical Nuclear Reactor Facility Modeling & Simulation Data—Scenario Comparison

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EXECUTIVE SUMMARY

This document evaluates a hypothetical nuclear power plant and associated protective force personnel using modern modeling and simulation tools. The facility incorporates a “Security by Design” methodology¹ to consider and integrate methods to resolve security issues and vulnerabilities via the facility’s inherent design characteristics before construction. The evaluation in this document is an example only. It is not intended to recommend or evaluate the effectiveness of existing physical security requirements or identify any method that the U.S. Nuclear Regulatory Commission staff may find acceptable for complying with existing requirements.

The facility, known as the “Small Modular Reactor Facility” or SMRF, is a state-of-the-art light-water reactor (LWR) nuclear plant designed to provide electricity to the fictitious city of Hashbakar. The facility houses four small modular reactors and various support buildings, including a vault to store category I nuclear material. This report provides a detailed description of the SMRF, including its reactor features, safety systems, and security measures, along with an analysis of potential threat scenarios and recommendations for maintaining and enhancing security.

The SMRF employs a comprehensive security strategy, including perimeter intrusion detection systems, redundant communication systems, and a well-trained onsite response team. The facility is protected by multiple layers of security, including access control points, surveillance cameras, and physical barriers. The security measures are designed to prevent unauthorized access and respond effectively to potential threats. Four adversary scenarios, modified from the Electric Power Research Institute’s risk analysis and management for critical asset protection methodology², were developed to evaluate the effectiveness of the SMRF’s security measures using modeling and simulation software:

Scenario 1—Targeted the special nuclear material vault, where an adversary force attempts to steal special nuclear material.

Scenario 2—Targeted the main control rooms, where an adversary force aims to infiltrate and sabotage the main control rooms.

Scenario 3—Targeted the auxiliary feedwater pumps, where an adversary force targets the auxiliary feedwater pumps.

¹ Snell, Mark K.; Jaeger, Cal; Jordan, Sabina E.; Scharmer, Carol S, “Security-by-design handbook,” SAND2013-0038, January 1, 2013, Japan Atomic Energy Agency, Ibaraki, Japan; Sandia National Laboratories, New Mexico.

² “Nuclear Power Plant Risk Analysis and Management for Critical Asset Protection (RAMCAP) Trial Applications Summary Report,” December 2005, Electric Power Research Institute; U.S. Department of Energy; American Society of Mechanical Engineers

Scenario 4—Targeted the auxiliary feedwater pumps; where an active adversary force, with support from an active insider, attempts to sabotage the auxiliary feedwater pumps.

These four scenarios were designed to test the physical protection system capabilities of the SMRF and associated response personnel against a fictitious design-basis threat. The results from analysis of these scenarios are each a probability of effectiveness for that scenario, which can be used to determine the effectiveness of the SMRF to defend against the fictitious design-basis threat in each of the modeled scenarios.

The methodology in this report determines the effectiveness of a facility with a high degree of security and the facility was rigorously tested through multiple adversary scenarios, including theft, sabotage, and insider threats.

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ACRONYMS

AC	Alternate Current
CAS	Central Alarm Station
CCTV	Closed-Circuit Television
CDP	Critical Detection Point
CFR	Code of Federal Regulations
DBT	Design-Basis Threat
ECCS	Emergency Core Cooling System
ECP	Entry Control Point
FoM	Figure of Merit
HEPA	High Efficiency Particulate Air; industrial standard
HVAC	Heating, Ventilation, & Air Conditioning
LAC	Last Access Control
LLEA	Local Law Enforcement Agency
LWR	Light-Water Reactor
MW _e	megawatt electric
MW _{th}	megawatt thermal
N/A	not applicable
OCA	Owner-Controlled Area
PA	Protected Area
P _A	Probability of Assessment
P _D	Probability of Detection
P _I	Probability of Interruption
PIDAS	Perimeter Intrusion Detection and Assessment System
P _E	Probability of Effectiveness

PIN	Personal Identification Number
P_N	Probability of Neutralization
PPS	Physical Protection System
P_s	Probability of Sensing
QRF	Quick Response Force
RF	Response Force
RTL	Response Team Lead
SAS	Secondary Alarm Station
SCIS	Safety Control and Instrumentation System
SDT	Scenario Development Team
SME	Subject Matter Expert
SMRF	Small Modular Reactor Facility

1 FACILITY CHARACTERIZATION

1.1 Description of Small Modular Reactor Facility

Note: The description of the facility's reactors is purely hypothetical.³

1.2 General Description

The Small Modular Reactor Facility (SMRF) is a fictitious nuclear facility, using a light-water reactor (LWR) design, with security and safety designed into the facility. Four small modular reactor units and a single spent fuel pool (shared by all reactors) are located below grade (basement level), a key feature that provides enhanced security and safety. The SMRF operates 24 hours per day 365 days per year.

A general operating description and facility safety information are provided in this section. The general plant layout is shown in Figure 1 (see page 15). From an operational perspective, the most important buildings or systems at the facility are the following:

- Four Reactor Buildings, containing four nuclear reactors, each within containment
- One Auxiliary Building
- One Control Room
- One Nuclear Receiving Building (containing a Special Nuclear Material Vault)
- One Non-Nuclear Receiving Building
- Four Turbine Buildings
- One Office Building
- Central Alarm Station (CAS) (in Office Building)
- Secondary Alarm Station (SAS)

In addition, the facility has buildings at security access points:

- Prescreen Administrative Building with Badge Office
- Personnel Access Control Portal
- Vehicle Screening Access Point

1.3 Reactor Features and Characteristics

Each of the four reactors uses natural circulation to cool the reactor upon shutdown. Each reactor module in the SMRF can produce 140 MW_{th} or 49 MW_e. As a plant of four reactor

³ Some characteristics retrieved from Tom G. Lewis, Benjamin B. Cipiti, Sabina E. Jordan, Gregory A. Baum, "Generic Small Modular Reactor Plant Design," December 2012, SAND2012-10406, and Alan S. Evans, Jordan M. Parks, Steven Horowitz, Luke Gilbert, Ryan Whalen, "U.S. Domestic Small Modular Reactor Security by Design," March 2021, SAND2021-0768.

modules, the SMRF can produce 196 MW_e. The following is a list of features and characteristics related to each SMRF nuclear reactor:

- The nuclear reactor core produces 140 MW_{th}.
- Each turbine is capable of producing 49 MW_e.
- A total of 39 fuel assemblies are arranged in a 17'17 rod bundle (typical of a pressurized-water reactor).
- The fuel is enriched to 19.9 % U-235, making the fuel high-assay low-enriched uranium.
- The primary cooling method is natural circulation.
- The reactor pressure vessel is 3 m in diameter and 10 m high.
- Interior confinement is 11.5 m in diameter and 16.5 m high.
- External diameter is 13.5 m and external height is 18.5 m (confinement thickness is just over 1 m).

1.4 Facility Buildings

Many of the buildings at the SMRF are constructed with at- or above-grade (ground-level or higher) and below-grade (basement) levels. The buildings at the SMRF are labeled on the facility layout (Figure 1).

1.4.1 Reactor Operations Buildings

The primary buildings for reactor operations are the following:

- The **Reactor Buildings** are all identical, reinforced concrete each approximately 100 by 100 feet.
- The **Auxiliary Building** (Figure 1) is 125 m long, 32.3 m wide, and 19.5 m high. The building has one ground floor and two basements. (see Figure 2, Figure 3, and Figure 4). This building houses both the spent fuel pool and an area for fresh and packaged fuel storage. For the purposes of this report, we will assume operating conditions where no spent fuel is being moved out of the spent fuel pool and the reactors are all operating at full power without spent fuel movement onsite.
 - The spent fuel pool is located in basement 2 and is accessible through the pool surface in basement 1. Hatches and cranes are used to transfer material, both nuclear and non-nuclear, from the ground floor to the basements. All building surfaces are constructed of 1 m thick reinforced concrete except for the floors, which are 0.5 m thick reinforced concrete.
 - Each nuclear reactor module contains two auxiliary feedwater pumps within the auxiliary building which feed water into the steam generator. Therefore, there are eight total auxiliary feedwater pumps. Only one of these pumps is necessary to sufficiently cool a module at a given time, but the modules are independent, so the two pumps can only cool the module they are connected to. If both pumps are damaged/disabled, without taking any mitigative or restorative measures, the

reactor core begins irreversible melting leading to a large radioactive release within six hours.

- Each **Control Room** is 22 m long, 19.7 m wide, and 9 m high. The control rooms are constructed of 1 m thick reinforced concrete, except for the floors, which are 0.5 m thick reinforced concrete.
 - The control rooms for all four reactor cores are located on the second floor of the main office building in the protected area (PA). The cable spreading room is located adjacent to the control room. Backup power sources for control room operations are in the ground floor of the auxiliary building, along with the heating, ventilation, and air conditioning (HVAC) units that provide separate HVAC in the control room. Two control room operators are always present in each control room.

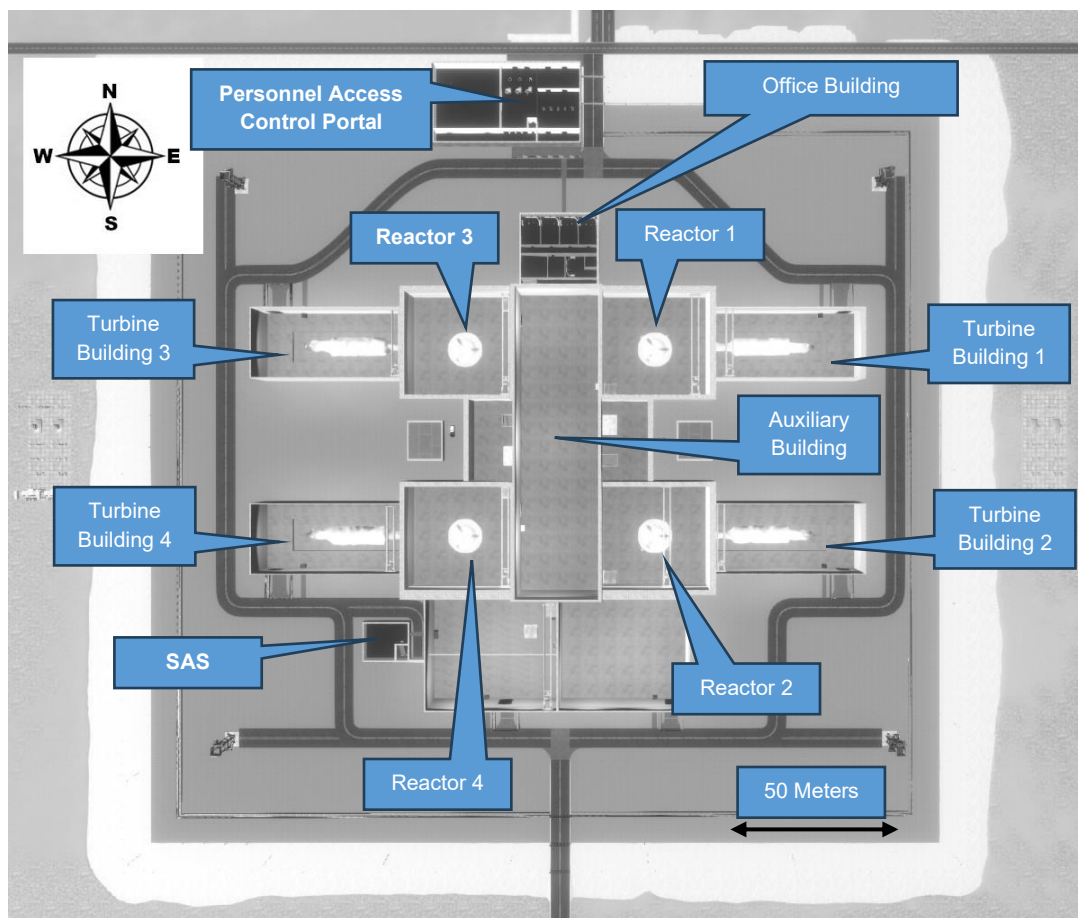


Figure 1. Small Modular Reactor Facility (SMRF) buildings with 50 meter scale

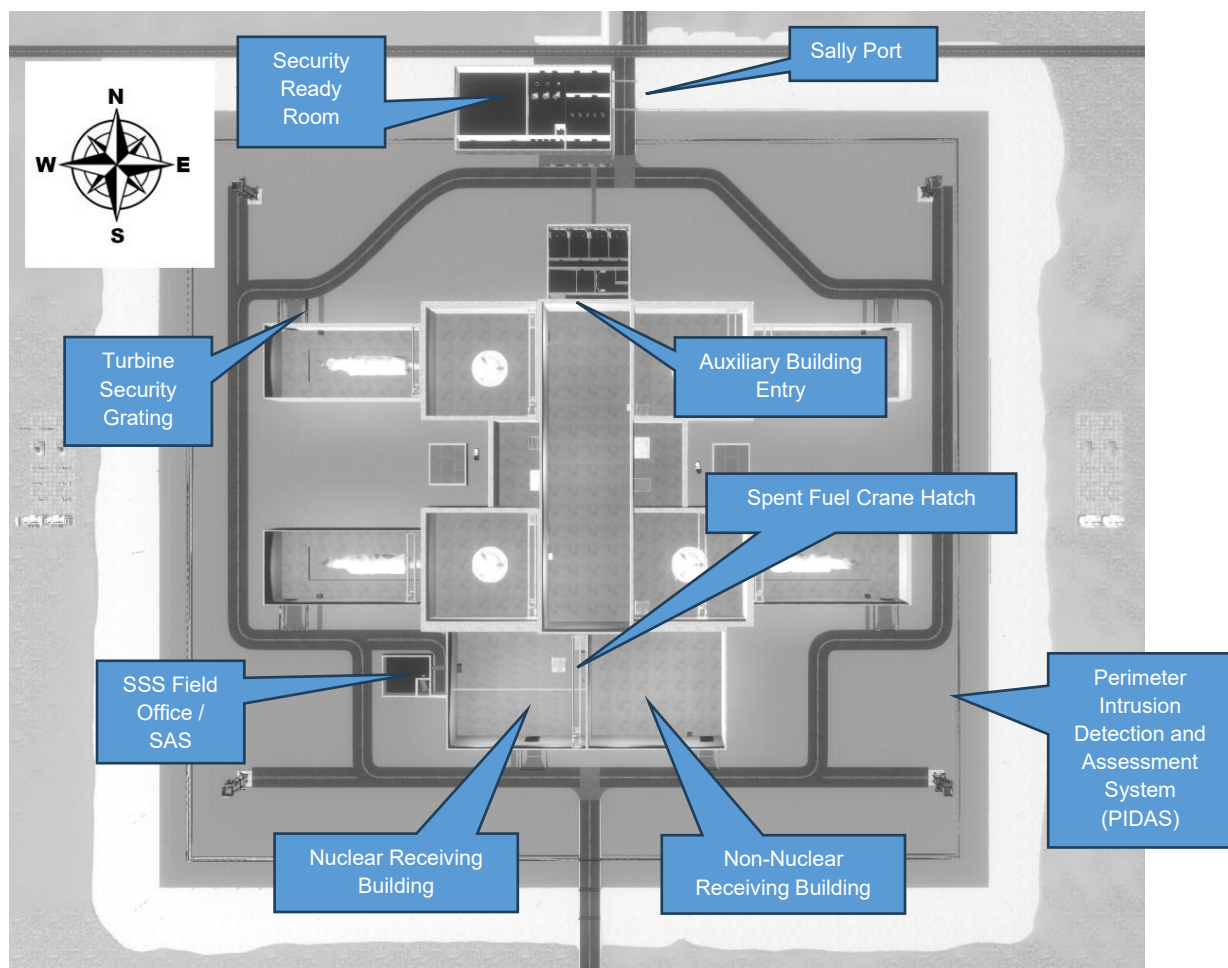


Figure 2. SMRF ground floor

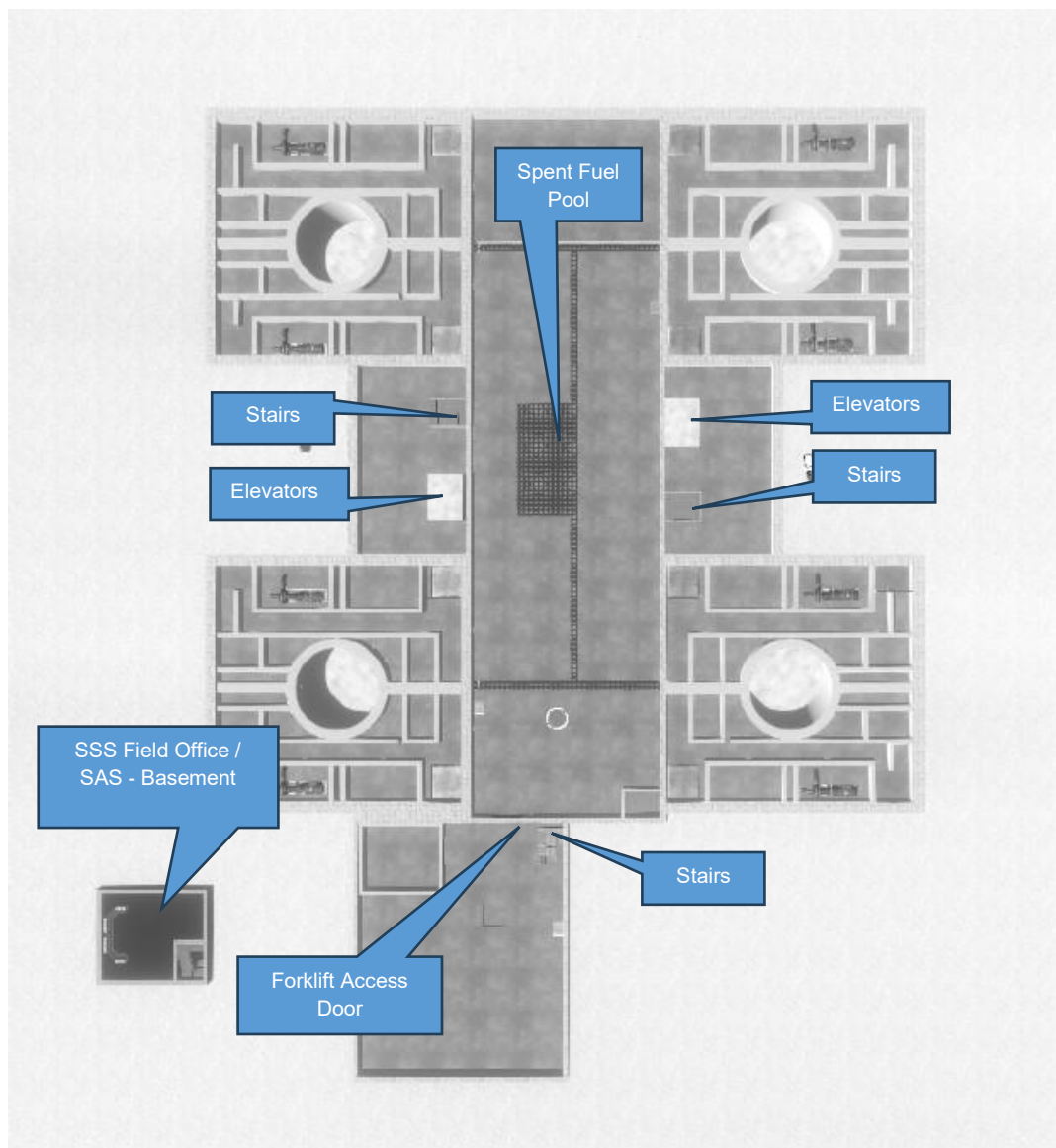


Figure 3. SMRF basement 1

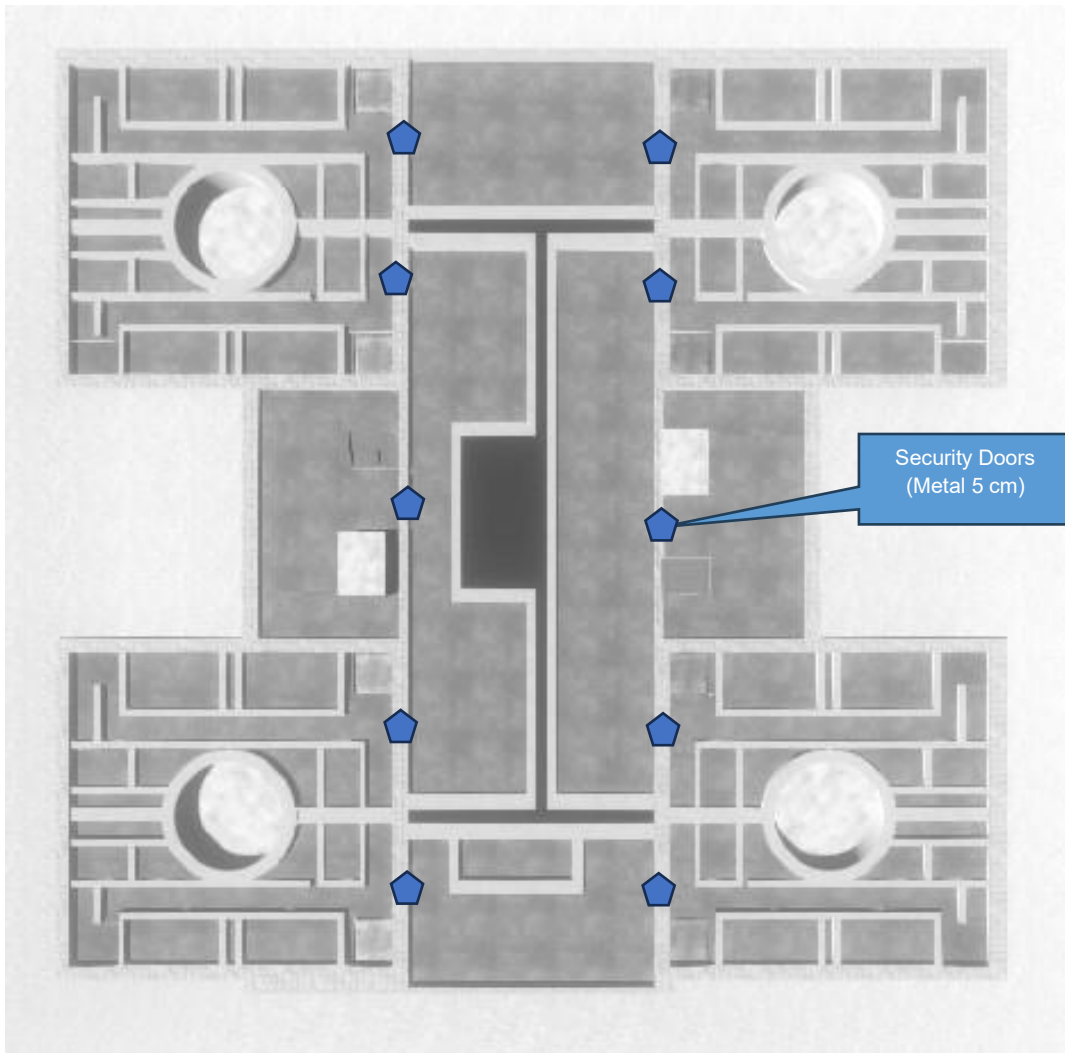


Figure 4. SMRF basement 2

- The **Nuclear Receiving Building** (Figure 1) is 46 m long, 37.2 m wide, and 5.5 m high. This building is the only entry and exit point for nuclear material at the facility. The building has one ground floor and one basement (see Figure 2 and Figure 3); it is not a seismic category I building.
 - The building contains the special nuclear material (SNM) vault which houses a category I amount of SNM. All building surfaces are constructed of 1 m thick reinforced concrete, except for the floors, which are 0.5 m thick reinforced concrete.
 - All material in the nuclear receiving building either is stored in casks or poses no significant radiological risk to the environment or plant personnel. Cranes are located in both the ground floor and basement; see Section 1.6.
- The **Non-Nuclear Receiving Building** (Figure 1) is 46 m long, 37.2 m wide, and 5.5 m high. This building has only a ground floor (see Figure 2). The non-nuclear receiving building is the entry and exit point for all non-nuclear equipment and supplies; it is not a seismic category I building. All building surfaces are made of 1 m thick reinforced concrete, except for the floors, which are 0.5 m thick reinforced concrete.
- Each of the four **Turbine Buildings** (turbine building 1–4 on Figure 1) is 43 m long, 44.5 m wide, and 19.5 m high. Each building is a seismic category I structure with one ground floor. (see Figure 2 and Figure 3). All building surfaces are constructed of 1 m thick reinforced concrete except for the floors, which are 0.5 m thick reinforced concrete.
 - The facility's transformer/generator system has a failure rate of 1/40 per operation year. In addition to the transformer/generator system, each turbine building contains reserve tanks of water and a battery bank for backup power as described in Section 1.5.5.

1.4.2 Administrative and Security Buildings

- The **Entry Control Point (ECP)** is located at the north side of the facility in the personnel access control portal building. The building houses the individual responsible for the last access control (LAC) function and security offices. This building has a ground floor only and 1 m-thick reinforced concrete walls. The floor is 0.5 m thick reinforced concrete. (see Figure 5, page 31).
- The **Office Building**, located inside the PA, includes administrative offices, security ready room, CAS, and the main control rooms on the second floor. The office building has a ground floor and one upper floor. (see Figure 2 and Figure 3 and Figures Figure 14 and Figure 15). It is constructed of 1 m thick reinforced concrete, except for the floors, which are 0.5 m thick reinforced concrete.

- The **SAS** is located to the west of the nuclear receiving building. It is a two-story facility with one ground floor and one basement floor. Walls are constructed of 0.5 m thick reinforced concrete.

1.5 Safety-Related Plant Components

Safety-related plant components are described below.

1.5.1 Shielding Structure, Confinement, and Reactor Pressure Vessel

Each reactor core is housed in a reactor pressure vessel, which is approximately 10 m high and 3 m in diameter. The pressure vessel sits inside a reactor confinement vessel that is approximately 18.5 m high and 13.5 m in diameter.

The reactor pressure vessel houses the reactor core, control rods, and steam generator. The reactor confinement vessel provides a layer that: (a) protects the reactor from external impact, (b) acts as a barrier to contain radioactive material, and (c) serves as the emergency cooling layer during decay heat removal of the reactor core upon shutdown. The confinement is filled with an inert gas to limit risk of fire.

Both the reactor modules and the spent fuel pool have an interior lining of 2 cm stainless steel for corrosion resistance. The spent fuel pool is open at the top to increase natural circulation to cool the spent fuel pool. Each reactor module is housed in an American Society of Mechanical Engineers Section III, Class I, thick-walled cylindrical pressure vessel with a removable top head.

1.5.2 Fresh and Spent Fuel

Fresh fuel is received on the ground floor of the nuclear receiving building and is stored in the SNM vault. During refueling it is transferred via two cranes to the fresh fuel and packaged fuel storage area in basement 1 of the reactor building. Each reactor is completely refueled every two years by replacing the entire core.

Spent fuel is removed from a reactor every two years and transferred via crane to the spent fuel pool. Spent fuel remains in the pool for approximately five years before being placed in dry-cask storage and shipped offsite. The spent fuel pool is cooled via forced-flow heat exchangers that direct water to the dry cooling radiators (see below). After five years, the spent fuel is transferred from the spent fuel pool to the fresh fuel and packaged fuel storage area. The packaged fuel is then moved out of the area using the same method as fresh fuel receipt, except in reverse. Dry-cask storage is present near the SMRF facility but is a separate site with its own security posture.

1.5.3 Dry Cooling System

Dry cooling radiators are used to cool the reactors and spent fuel pool without access to water. Fans for the dry cooling radiators use offsite power to ensure that a power surge does not affect the facility's safe shutdown. In the case of offsite power loss, the dry cooling

radiator fans can be powered by backup emergency diesel generators. Placement of the dry cooling radiators is arranged to increase natural circulation of airflow; steel beams are positioned around the radiators to prevent direct vehicle assault.

1.5.4 Core Cooling System

The reactor core operates at a low power density compared to typical existing light-water power reactors, making natural circulation adequate for cooling the core during normal plant operations and design-basis accident scenarios. The core cooling system is a simple system, consisting of the primary core coolant flow passively driven by gravity, a closed secondary coolant loop, and a closed tertiary reactor coolant loop. The coolant within the primary system is kept in a state of liquid water.

In an integral pressurized water reactor design, the steam generator is located within the reactor pressure vessel. The core cooling system's secondary loop allows steam to be generated inside the reactor pressure vessel within the steam generator and passed to a turbine outside of reactor confinement. Remaining steam is then sent to the condenser, where it is ultimately condensed back into liquid water by a cooler source of water. The water returns to the steam generator within the reactor pressure vessel via forced circulation by two separate feedwater pumps per reactor unit, completing the secondary loop.

Water supplied to the condenser to cool the steam exiting the turbine is contained in the closed tertiary loop. This system supplies cool water to the condenser and is ultimately cooled using forced mechanical draft coolers. Each reactor is equipped with three mechanical draft coolers. Normal operation requires one fully operational cooler to ensure proper cooling. Therefore, at least five coolers (one for each reactor plus at least one for the spent fuel pool) must be operational.

Should primary onsite power be lost, all core cooling system equipment is run via battery power. The battery power is designed to provide two hours of emergency power to the core cooling system equipment. If loss of power lasts more than two hours, irreversible onset of core damage results at six hours after loss of onsite power.

1.5.5 Emergency Core Cooling System (ECCS)

Each reactor vessel is equipped with an ECCS. Each ECCS is five meters in diameter and eight meters high. The ECCS delivers water to the top of the confinement structure to cover the reactor pressure vessel with water. As the water boils off, water is continually added by the ECCS from the reserve water tanks located in the basement of the turbine buildings to ensure cooling of the reactor core. Alarms in the steam generator or the condenser automatically activate the ECCS.

In the event of a loss-of-coolant accident or low steam level in the steam generator, the ECCS will be activated automatically. The controllers in the main control rooms can also activate the ECCS.

When the ECCS is initiated, a sensor of the water level in the reactor confinement structure triggers the opening of a pneumatic valve, allowing water from the ECCS's ultimate heat sink to enter the reactor confinement vessel. When a predetermined water level is reached inside confinement, the pneumatic valve is automatically shut.

As heat is transferred to water entering confinement via the ECCS's ultimate heat sink, the water will evaporate. This water vapor passes through a high efficiency particulate air (HEPA) filter and then is vented out through the roof of the structure. The HEPA filter is designed to capture radionuclides in conjunction with a radioactive material detector positioned after the HEPA filter. The tracking data is reported to the control room, allowing controllers to discern if venting the steam from the reactor building is appropriate.

As water continues to evaporate inside confinement, the water level monitor inside confinement triggers the pneumatic valve to open so that water from the ultimate heat sink can continue to enter the reactor confinement structure.

Reserve Tanks. The turbine building also houses reserve ECCS tanks to ensure the cooling of the reactor cores, providing the following:

- Chemical volume control system water, which maintains water quality for the reactor pressure vessel.
- Borated water, which can be sent directly into the reactor pressure vessel through the high-pressure safety injection system pumps.

Backup Power. Should the SMRF lose offsite primary power, backup battery banks located in the auxiliary building would allow the ECCS to operate. If onsite primary power were lost, then all four reactor cores begin a shutdown procedure (i.e., a "SCRAM") and the ECCS system automatically activates. The transition time from primary power to backup battery power takes approximately 30 seconds. Battery power is needed to operate the pneumatic valve and water level indicator in the confinement structure during primary power loss. Battery power can sustain core cooling for two hours before irreversible core damage initiates at six hours after loss of onsite power.

1.5.6 Other Safety-Related Features and Systems

- **Switchyard.** The facility has two switchyards for transmission, distribution, collection, and control of the flow of electricity. Each switchyard is subdivided into two separate subyards. Each subyard is equipped with access controls and serves one reactor. The switchyards, located in the limited access area, are surrounded by a three-meter-high chain-link fence, and topped with concertina wire.
- **Safety Control and Instrumentation System (SCIS).** This system is part of the instrumentation and controls systems. The SCIS is charged with ensuring the control of reactivity in the reactor core, monitoring the removal of heat from the reactor core, and monitoring confinement or any release of radioactive material.

- **Alternate Current (AC) Power.** An AC power distribution system provides power to the plant for all normal operations, including startup, operation, shutdown, and outage operations. In the case of power loss, a safety set of backup diesel generators starts automatically. The safety generators are located in the auxiliary building behind watertight doors that protect them from flooding. The diesel generators are small enough that air cooling is adequate. Air cooling increases system reliability.
- **Reactivity Control System.** Reactivity control is primarily achieved through electrically driven control rods in what is known as the reactor protection system. This system uses the fine control rod drives to insert control rods and remove control rods from the reactor core. In a reactor-controlled reactor protection system shutdown (i.e., a “SCRAM”), the drives quickly insert the control rods into the core and shut down the reactor.
- **Standby Liquid Control System.** If the controller cannot ensure reactor shutdown through the reactor protection system, the standby liquid control system can be used to flood the reactor core with borated water. The system is activated through automatic alarms or by controller action. The borated water is injected through a high-pressure line. The pressure is high enough to overcome any theoretical reactor condition.
- **Safety-Related HVAC.** The HVAC emergency cooling water system provides chilled water to the safety-related equipment in the reactor building and control rooms. The system is designed to work under normal and abnormal reactor conditions. In the event of abnormal conditions, the system can be powered by other AC power sources. The facility design provides some protection from missiles to the system. The system is also protected from abnormal power supply surges and short transient operations. Fill tanks and associated equipment are designed to prevent drainage through operation.
- **Primary Confinement Service Air System.** This system, which is rated as seismic category I, is designed to redundantly ensure primary air quality for normal operation.
- **Fire Confinement/Control System.** Reactor safety shutdown equipment has redundant systems, each capable of shutting down the reactor safely. Each system is separated by firewalls to prevent any single fire from removing the redundant safety systems. Fire suppression is provided by sprinklers, an aqueous film forming foam sprinkler system, alarms, detectors, portable firefighting equipment, and other expected fire suppression systems. Fire in confinement is not possible during operation due to the inert nature of confinement. Special fire prevention/suppression measures are taken when confinement is not inert.
- **Communication Equipment.** The SMRF has access to a powered telephone line for communications to offsite emergency agencies for assistance. The SMRF also has a sound-powered telephone system used in safety-related areas of the facility.

1.6 Operating Procedures

1.6.1 Fuel Shipments, Storage, and Packaging

Fuel shipments are received at the nuclear receiving building, having passed through the owner-controlled area (OCA) and PA vehicle portals.

Fresh Fuel Transfer. Fresh fuel is received on the ground floor of the nuclear receiving building (see Figure 2) and transferred via crane (crane 1) to basement 1. In the basement, the fuel is attached to another crane (crane 2) so that the fresh fuel can be moved into the fresh fuel and packaged fuel storage area (see Figure 2 and Figure 3). Both crane 1 and crane 2 have set weight limits so that no more than one fuel assembly or cask can be moved at a time. The controllers set the weight limits for specific movement of material, and the cranes operate at a speed of one meter per second. In addition, a third crane (crane 3) transfers fuel to a reactor. Each reactor is completely refueled every two years by replacing the entire core (see Section 1.6.2). Note: The crane icon in the fresh fuel and packaged fuel storage area (Figure 3) represents the location of both crane 2 and crane 3.

Spent Fuel Transfer and Packaging. Spent fuel is removed from a reactor via crane 3 and transferred to the spent fuel pool in basement 1 of the reactor building (Figure 2). Spent fuel remains in the pool for approximately five years. When it is time to ship, the spent fuel is moved out of the facility in reverse order of fresh fuel receipt; crane 3 is used to transfer the spent fuel from the pool to the fresh fuel and packaged fuel storage area so it can be packaged. Crane 2 is then used to transfer the packaged fuel to basement 1 of the nuclear receiving building. The spent fuel is attached there to crane 1 for transfer to the ground floor of the nuclear receiving building.

Storage. The fresh fuel and packaged fuel storage area in basement 1 of the reactor building provides storage for fresh fuel and a packaging area for spent fuel.

1.6.2 Refueling

Each reactor is completely refueled by replacing the entire core every two years. Reactors are refueled via cranes. The refueling process takes approximately three weeks, 24 hours per day. The refueling process includes the following steps:

1. Shut down reactor, turbine, and generator.
2. Cool down the reactor, depressurize, and allow for flood confinement.
3. Remove shield plugs.
4. Prepare atmosphere in shielding structure.
5. Remove and plug confinement/reactor plugs.
6. Remove confinement insulation.
7. Remove reactor pressure vessel head.
8. Remove steam generator and other internals.
9. Remove spent core.
10. Insert new core.

11. Verify new core arrangement.
12. Inspect vessel.
13. Inspect internals.
14. Inspect steam generator.
15. Inspect control rods.
16. Replace detectors.
17. Reverse dismantling process.

2 THREAT CHARACTERIZATION

The NRC defines its two design-basis threats (DBTs): one for facilities containing category I SNM for theft and the other radiological sabotage of commercial nuclear power plants. Both are described at an unclassified, public level in 10 Code of Federal Regulations (CFR) Part 73. Specific characteristics of both of these threats are withheld from public disclosure as sensitive information. For the purposes of this report, a fictitious threat profile was developed. This threat profile, while not reflective of the NRC DBTs, includes the number of adversaries, their capabilities, and equipment.

As this fictitious threat profile is a stand-in for the DBT, the development of the threat profile is not a part of this process. Instead, the threat was created separately. In this step, the threat and its capabilities were incorporated into the analysis.

3 SMALL MODULAR REACTOR FACILITY TARGET IDENTIFICATION

3.1 Theft Targets

Table 1. Theft target including nuclear materials and enrichment

Building	Location	Form of Material	Amount of Material on Site (wt% Enrichment)	Total Isotope Amounts	Level of Radiation
Nuclear receiving building	SNM vault	UO ₂ pellets in rods	10 kg U (80 % U-235)	8 kg U-235	Low

3.2 Sabotage Targets

Note: All scenarios and targets are fictitious.

Once fuel has been used in the SMRF, it produces high levels of radiation and heat due to the decay of the fission products. Current NRC requirements for commercial LWRs under Title 10 CFR Part 73.55⁴ requires the prevention of significant core damage and spent fuel sabotage. Two new proposed security-related rulemakings propose to change the criteria for prevention of radiological sabotage to “the prevention of a significant release of radionuclides from any source.”⁵

Commercial nuclear reactor facilities should be analyzed according to Regulatory Guide 5.81, Revision 1 or Revision 2⁶ to identify the target sets that if compromised would lead to irreversible core damage or a release of radionuclides from any source, depending on which regulatory framework the applicant is under. These sabotage targets are outlined in

⁴ PART 73—PHYSICAL PROTECTION OF PLANTS AND MATERIALS

⁵ For the alternative physical security requirements for advanced reactors proposed limited-scope physical security rulemaking, proposing to amend the requirements of existing 10 CFR Parts 50 and 52, please see NRC-2017-0227, <https://www.regulations.gov/docket/NRC-2017-0227>. For the NRC’s proposed risk-informed technology-inclusive regulatory framework for advanced reactors, please see NRC-2019-0062, <https://www.regulations.gov/docket/NRC-2019-0062>.

⁶ Target set identification and development for nuclear power plants, Regulatory Guide 5.81, December 2019, Revision 1 or the forthcoming Rev. 2 Draft Regulatory Guide DG 5071 [forthcoming].

Table 2.

Table 2. Sabotage targets

Building	Location	Form of Material	Level of Radiation
Reactor building	Reactor cores	Irradiated fuel	High
Auxiliary building	Spent fuel pool	Spent fuel	High
Auxiliary building	Auxiliary building	N/A	Medium
Control room	Office building	N/A	Low

4 RESPONSE FORCE CHARACTERIZATION

4.1 Response Force Personnel

The response force (RF) consists of an onsite armed security RF with 26 armed members per shift.

4.1.1 Security Officers

The armed officers are responsible for:

- Assessment of alarms.
- Administrative duties such as access control, visitor badge issuance, and key service.
- Routine patrols and staffing of fixed posts.
- Response to all intrusion alarms.
- Monitoring nuclear material transfers with nuclear material control and accounting personnel.
- Personnel, package, and vehicle searches.

All posts and patrols have defined policies and procedures with which the officers must comply.

4.1.2 Security Supervisors

For each shift, two security supervisors are present at the facility. These positions supervise the day-to-day regulatory required operations and serve as command and control for all contingency events.

4.2 Equipment

4.2.1 Officers

All officers are equipped with:

- Oleoresin capsicum spray
- One set of handcuffs
- Flashlight
- Handheld radio
- Response weapons (5.56 rifle & 9mm Glock 17)
- Ammunition (150 rounds 5.56 & 52 rounds 9mm)
- Keys
- Response vest
- Gas mask

4.3 **Response Force Training**

4.3.1 **Classroom Training (All Security Staff)**

All officers are trained on the following:

- Access control procedures
- Use of force continuum
- Target locations
- Response procedures
- Chain of command
- Other administrative responsibilities
- Close quarters combat
- Recapture and recovery of nuclear material/facilities
- Advanced firearms training for both the pistol and the assault rifle
- Firearms training each month to ensure proficiency

All personnel receive routine physical fitness training when in training mode.

4.4 **Deployment of Response Force**

The RF is deployed as described in Table 3 and shown on the site layout (Figure 5). An X in regulatory space refers to a responder with a dedicated timeline and initial response position. A Y is a responder that has no timeline and does not have a dedicated initial response position.

Table 3. Response force deployment data

Responder	Armed	Unarmed	X	Y	Supervisor	Call Signs
ECP/vehicle barrier system	1	1 (does not fit into categories)		1		Echo 2
Quick Response Force (QRF) 1 (north)	6		2	4		November 1–6
LAC	1			1		Lima 1
QRF 2 (south)	6			6		Sierra 1–6
Mike 1 & 1A	2			2		Mike 1 & 1A
Mike 2 & 2A	2			2		Mike 2 & 2A
Response Team Lead (RTL)	1			1		Romeo 1
Site Security Manager		1			1	Charlie 1
Blast & Ballistic Rated Enclosure	4		4			Bravo 1–4
Field Supervisor	1			1		Foxtrot 1
CAS/SAS	2				2	CAS/SAS

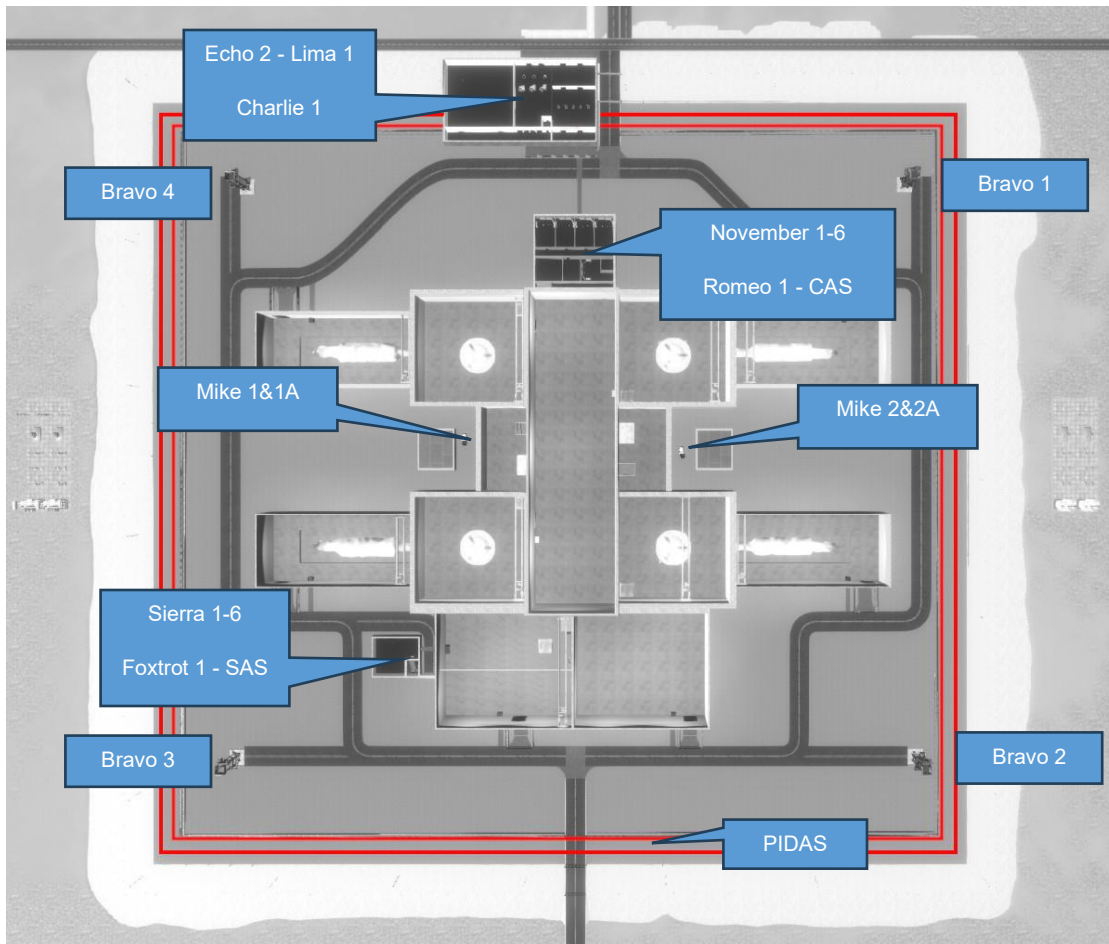


Figure 5. Overall facility layout with response posts and perimeter intrusion detection and assessment system (PIDAS)

4.5 Response Procedure

All alarms are received simultaneously at the CAS and SAS. When an alarm is assessed as a malicious intrusion, CAS will immediately notify the onsite security RF to begin deployment. Bravo 1–Bravo 4 will make their weapons ready and engage any hostile threats. November 1 and November 2 will deploy to a defensive fighting position outside of the security ready room and provide protection for pathways leading to the auxiliary building, control rooms, and CAS. Sierra 1–Sierra 6 will be deployed by the RTL at their discretion inside the auxiliary building. November 3–November 6 can be deployed by the RTL depending on the direction of the attack once they lock down the ECP.

Upon implementing the physical protection strategy, CAS then notifies the control rooms of a hostile action at the facility. Once the control rooms have been notified, contact with the local law enforcement agencies (LLEA) will be made.

4.6 Response Force Performance Data

The facility has conducted extensive performance testing of the CAS in the areas of alarm assessment, alarm communication, and RF notification. The average times are listed in Table 4.

Table 4. Average response times

Description	Reactor Time (seconds)
Alarm communication time	15
Alarm assessment time	30
Communication time to onsite responders	10
Communication time to the control rooms	10
Communication time to LLEA	30
Officer preparation time	8
Officer travel and deployment time	10

4.7 Alarm Station Staffing, Locations, and Communications

The CAS and SAS are staffed with at least one officer each at all times. The designated RTL resides in the CAS.

The CAS, which is located on the second floor of the main office building, has solid reinforced masonry walls and no windows. The door to the CAS is a five-centimeter-thick metal security door, secured by both a magnetic lock and an electromechanical bolt. The door can be opened only by an authorized user (with a proximity badge) along with permission of the CAS operator, who releases the electromechanical bolt. The operator confirms the authority of the person requesting entry through a video intercom.

The CAS is equipped with an automated digital alarm and video annunciation and assessment system. All alarms, door access notifications, and video cameras are routed to the CAS on supervised signal wires within metal conduit. The alarms annunciate through a personal-computer-based graphical interface, and alarms are displayed along with a text description on a detailed schematic of the facility. The video camera recording—10 seconds before and after the alarm—is displayed on loop automatically so that the operator can quickly assess the cause of the alarm. Live video feed from the nearest closed-circuit television (CCTV) camera to the alarm is automatically displayed as well. The alarm communications and display software prompts the operator to acknowledge incoming alarms and provides a checklist of actions to be performed. Operators can manually select and observe any camera to provide surveillance of activities.

The CAS includes redundant communication systems to communicate with onsite officers, offsite LLEA, and safety responders. Communications include a dispatch telephone station

and base radio station. Portable radio sets are provided to security posts and to security officers on post and patrol. All alarm assessment and communication capabilities can be conducted in the SAS as well.

When an alarm is received but the CAS cannot confirm the cause, the operator communicates the alarm to the perimeter security towers. The closest security tower will visually investigate the received alarm. Upon assessment of the alarm, the CAS will either clear the alarm or announce a security condition.

Duress Switch. In the case of duress, the operator can press a duress button to alert all officers of an issue.

Communications System. Extensive testing of the communications system has shown that radio communications are effective throughout the facility. However, it has been determined that officers inside reactor building basement 1 are able to monitor transmissions from the CAS but unable to transmit with their handheld radios.

4.8 Prescreen Building and Owner-Controlled Area (OCA)

The SMRF facility is inside an OCA.

4.8.1 Prescreen Building

The prescreen building at the entrance to the OCA provides visitors with facility information. The badge office and other administration offices are housed in the prescreen building.

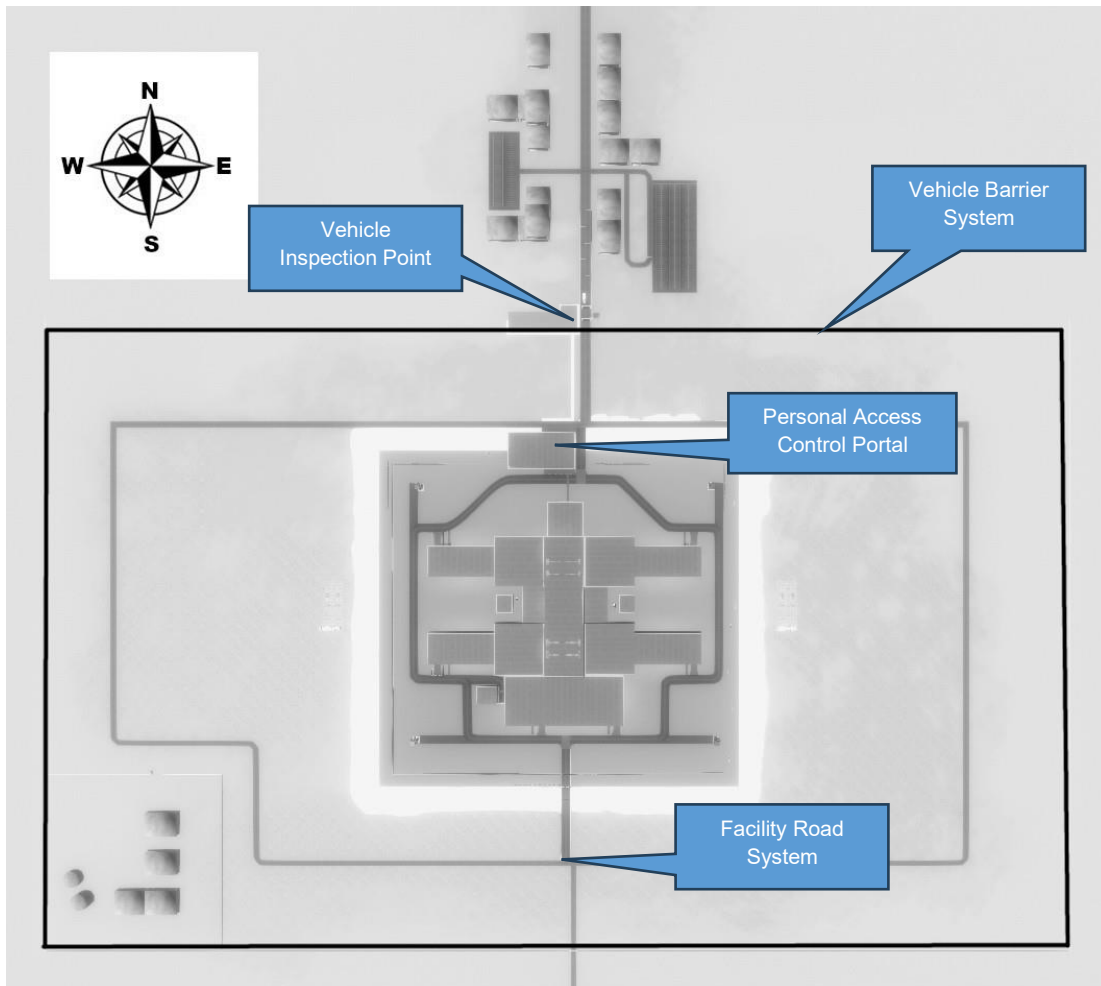


Figure 6. OCA boundary and access control points

4.9 OCA Protection Elements

The OCA is bordered by a concrete vehicle barrier system with a vehicle entry/exit portal.

4.9.1 OCA Perimeter

The OCA is surrounded by a continuous poured concrete wall, with sections comprised of modular concrete blocks for larger equipment entry if needed. The wall is reinforced with inch-thick steel rebar and measures 5 ft tall by 12 ft wide.

4.9.2 OCA Entry Locations

- **Personnel Portal.** The entry point located inside the prescreen building is the only access control point that allows pedestrian entry into the OCA.

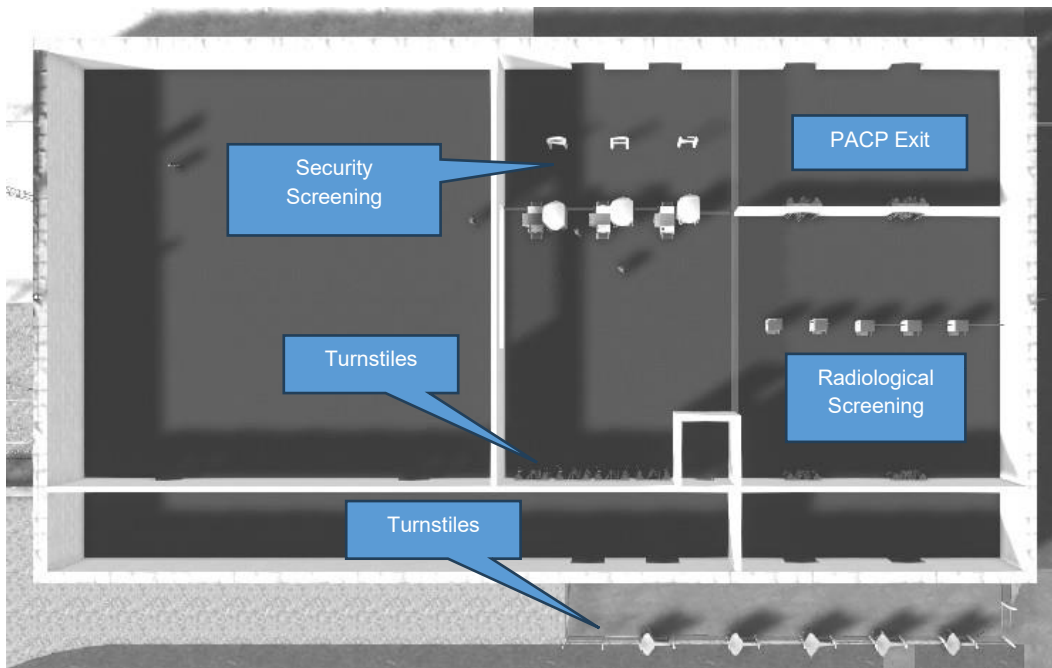


Figure 7. Personnel access control portal

- **Vehicle Portal.** The automated hydraulic wedge barriers can only be operated by the LAC officer located in the entry control building. This gate is the only vehicle entry point into the OCA. The barriers are normally locked and alarmed when the post is not staffed.

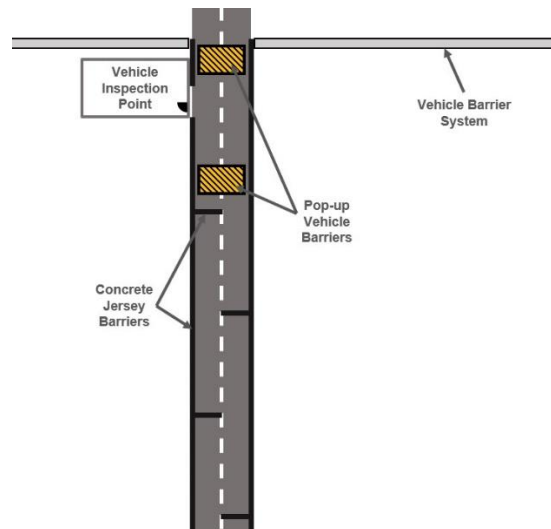


Figure 8. Vehicle access control portal

4.10 Protected Area (PA) Protection Elements

4.10.1 PA Perimeter

The PA, which is shown in Figure 9, includes all operational facilities at the SMRF.

4.10.2 Perimeter Intrusion Detection and Assessment System (PIDAS)

The PIDAS around the PA includes the following protection elements:

- Two fences separated by an isolation zone.
- Bistatic microwaves in the isolation zone to detect unauthorized access.

4.10.3 Perimeter Cameras and Lighting

Fixed cameras are installed around the perimeter at approximately 50 m intervals. Lighting includes the following features:

- 1000 W incandescent floodlights are mounted on poles at 10 m high.
- The surface is low reflectivity dark gravel with some vegetation.
- Lights are activated by exterior infrared sensors.

4.10.4 PA Entry Control Point (ECP) Locations, Staffing, and Process

The ECP on the northern boundary of the PA provides access into the PA. The portal is located at the perimeter and its roof is alarmed to complement the detection coverage of the intrusion detection system.

Both the personnel and vehicle portals are monitored by CCTV cameras and can be electronically locked on a time delay. All access control in the PA is monitored for personnel entering and exiting the buildings for emergency preparedness and security.

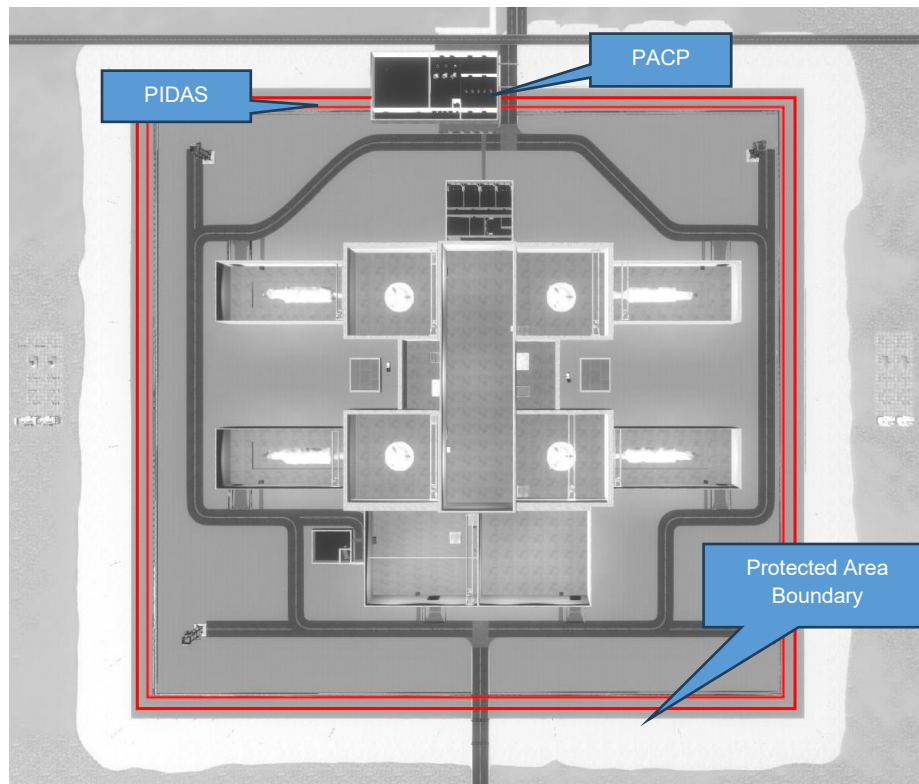


Figure 9. PA boundary and access control points

4.10.5 Pedestrian Entry/Exit

All personnel, including security team members, enter the PA through the ECP using a badge swipe, personal identification number (PIN), and retinal scan, see Figure 7. Two officers are posted at the personnel portal at all times.

4.10.6 Vehicle Entry/Exit

The vehicle portal into the PA includes inbound and outbound reinforced gates, as shown in Figure 10. One officer is always posted at this access control point during operational hours.

Note: Only authorized facility vehicles and approved vendors are allowed access into the PA.

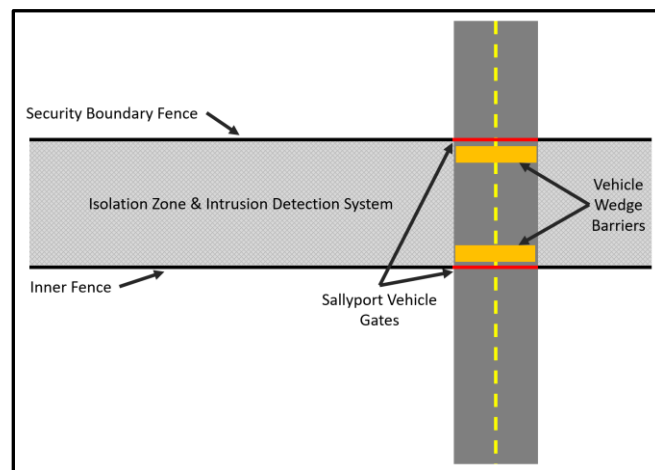


Figure 10. Entry into vehicle access control portal

4.11 General

All personnel access is monitored and logged for entering and exiting the buildings in the PA for the purpose of emergency preparedness and security.

Access to Ground Floors. Access to the ground floor of every building requires authorization (proximity badge, PIN, and retinal scan).

Exterior access for personnel is provided at the office building, non-nuclear receiving building, nuclear receiving building, and turbine buildings, see Figure 2. No officers are posted at any of these access points.

All exterior and interior doors in the PA are five-centimeter-thick metal security doors.

Access to Basement 1 at Nuclear Receiving Building. The basement of the nuclear receiving building is accessible to personnel. After entering the building by means of the

exterior door, personnel can take the stairwell to the building's basement, see Figure 11. In the event of a malicious act, the stairwell door can be locked.

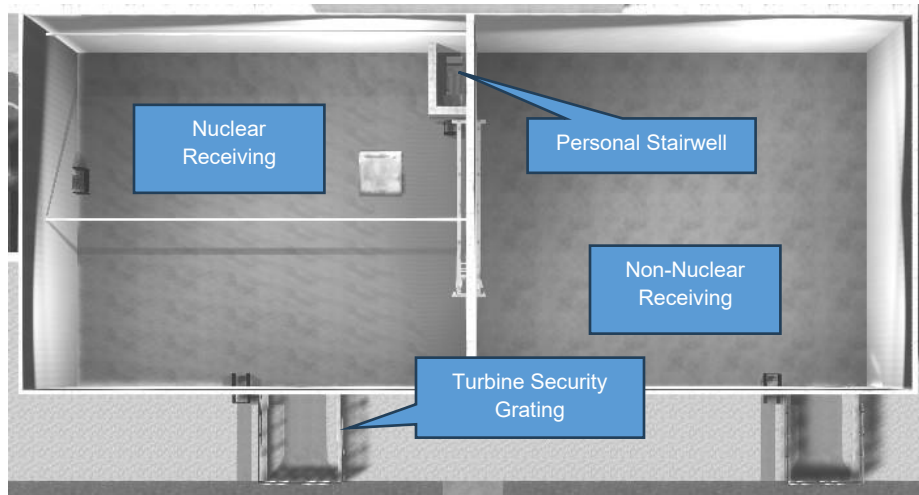


Figure 11. Nuclear receiving building access

Overview of Controls. Table 5 provides an overview of physical barriers and access controls.

Table 5. Small Modular Reactor Facility building access controls and physical barriers

Building	Plant Area	Equipment of Interest	Access Controls and Physical Barriers
Reactor building	Shielding structure	Confinement and safety-related equipment	Proximity badge access control with radiation interlock and PIN verification. Earthquake-resistant, bullet-resistant barriers and doors
	Ultimate heat sink (4)	Water required for passive safety	Double-wall tank, 24-inch concrete wall, and earthquake-resistant barrier. Lock-and-key access control to valves, and other insider sabotage targets
Control rooms	Control rooms	Controls for plant and passive safety systems	Entry and exit procedures with proximity badge access, PIN, and retinal scan verification. Earthquake-resistant building design
	Cable spreading room	Cables for control of plant operating systems and engineered safety features	Proximity badge access control
Nuclear receiving building	Vehicle portal	Entry controls	Entry and exit procedures with proximity badge access, PIN, and retinal scan verification. Earthquake-resistant building design. Turbine grating delay barriers
	Personnel door	Entry controls	Entry and exit procedures with proximity badge access, PIN, and retinal scan verification. Shark cage on the exterior
Turbine building 1 – 4	Vehicle portal	Plant capital protection equipment	Entry and exit procedures with proximity badge access, PIN, and retinal scan verification. Earthquake-resistant building design. Turbine grating delay barriers
	Personnel door	Plant capital protection equipment	Entry and exit procedures with proximity badge access, PIN, and retinal scan verification. Shark cage on the exterior

4.12 Specific Building Access

Personnel can gain access to the **Reactor Building's** ground floor and basement. Access to the ground floor is from the non-nuclear receiving building vehicle door and/or personnel door, see Figure 2. Access to reactor building basement 1 is either through the office building's personnel door or through the nuclear receiving building's basement vehicle and/or personnel door, see Figure 3. In typical day-to-day operations, personnel must enter through the security door in the office building using a badge swipe, PIN verification, and retinal scan.

All access control points are monitored from inside the central alarm station. Access to the inside of the reactor can also be gained through the crane 2 hatch for items. An officer must be present when the hatch is open. Opening the hatches initiates a reactor shutdown if the reactor is in operating mode, which triggers an immediate alarm in the central and secondary alarm stations. The hatch may be opened only from the control room.

The **Control Rooms** are accessible only through the office building. To reach the control rooms, personnel must process through the security door on the ground level and then take the stairs to the second floor where they will have to process through another security door. Once they have processed through the second level security door, they must process through another security door which leads into the control rooms. A proximity badge, PIN, and retinal scan verification are required to access all doors leading to the control room.

The **Nuclear Receiving Building** is the only vehicle entry and exit point for nuclear material. The ground floor has one access control point into basement 1. A proximity badge, PIN, and retinal scan verification are required for access into the building. Fresh fuel is moved from the ground floor of the nuclear receiving building to basement 1 via crane 1. Once in the basement of the nuclear receiving building, the fuel is moved to the reactor building basement 1 by crane 2.

In the reactor building, personnel use crane 2 to move the fuel into the fresh fuel and packaged fuel storage room. Personnel can enter the basement through the stairs from the nuclear receiving building's ground floor; personnel then enter the reactor building through the personnel door to detach the fuel from the crane and store the fuel in the fresh fuel and packaged fuel storage area. A proximity badge, PIN, and retinal scan verification are required for access to the door into the reactor building and storage area; access is monitored continually by the CAS.

The **Non-Nuclear Receiving Building**, which has a ground floor only, is the entry and exit point for all non-nuclear equipment and supplies. Figure 2 illustrates ground floor access to the reactor building. Access is through the personnel door or, for large equipment, the vehicle door. Access requires proximity badge, PIN, and retinal scan verification.

Access into a **Turbine Building** requires proximity badge, PIN, and retinal scan verification. Figure 2 illustrates building ground floor access to the reactor building.

4.13 Interior Sensors and Cameras

All interior doors are hardened but were not locked in this simulation. They include balanced magnetic switch sensors and are monitored by fixed cameras. Figure 12 through Figure 19 show placement of physical protection system (PPS) elements on the ground floor.

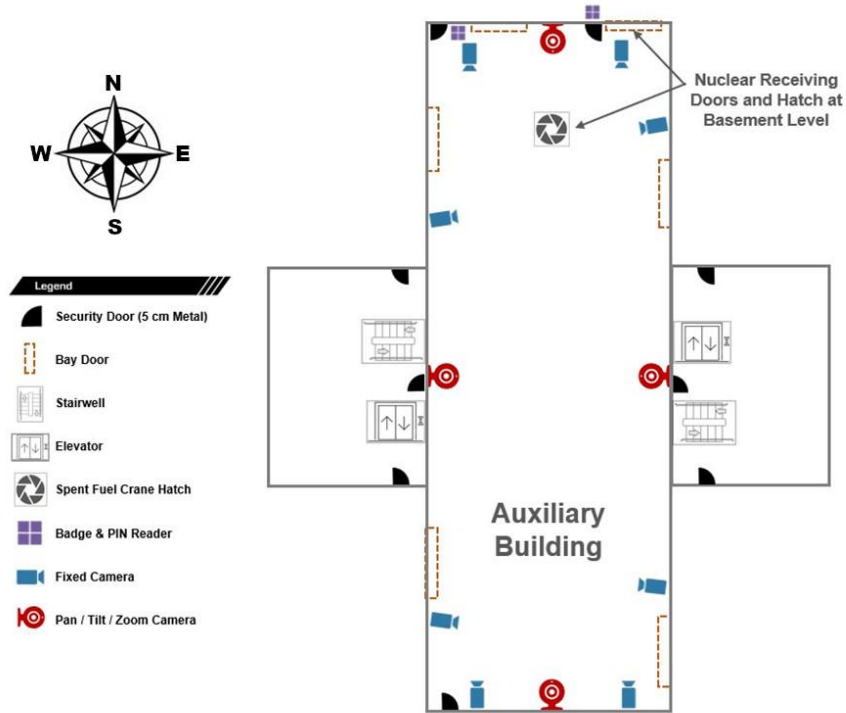


Figure 12. Physical security elements in reactor building, ground floor

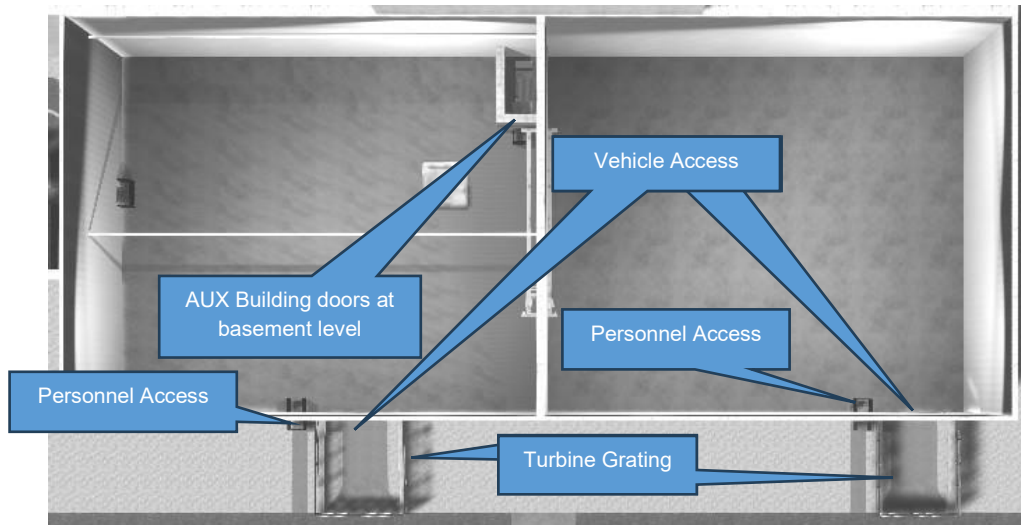


Figure 13. Physical security elements in receiving buildings, ground floor

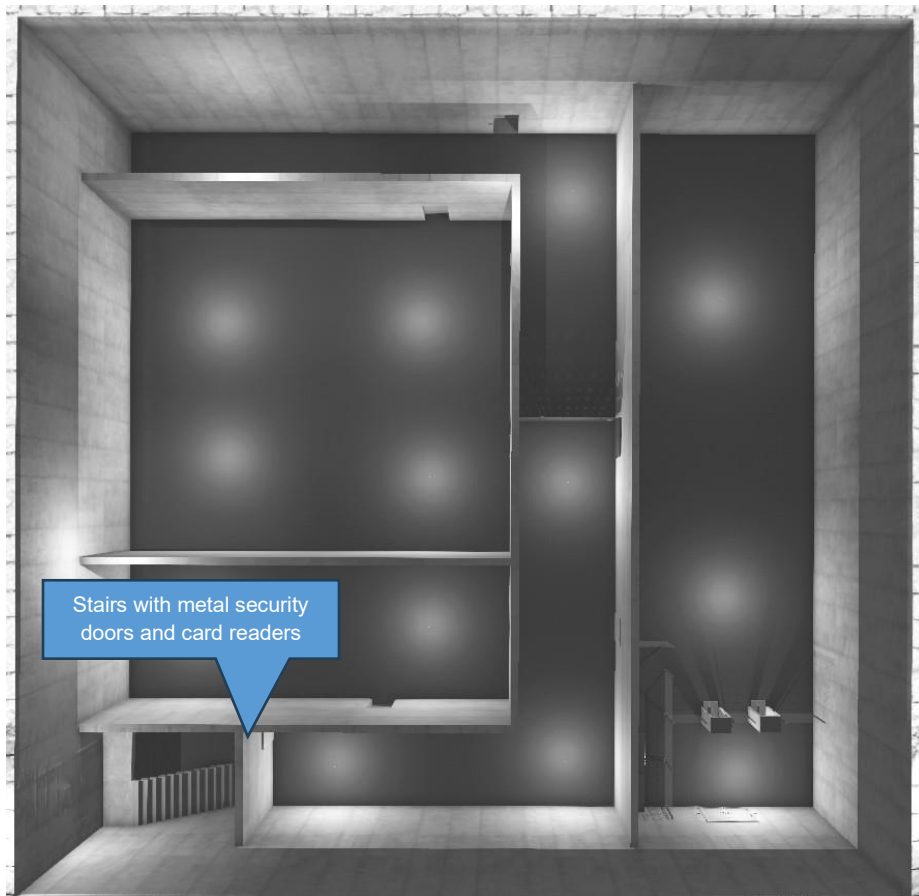


Figure 14. Physical security elements in office building, ground floor

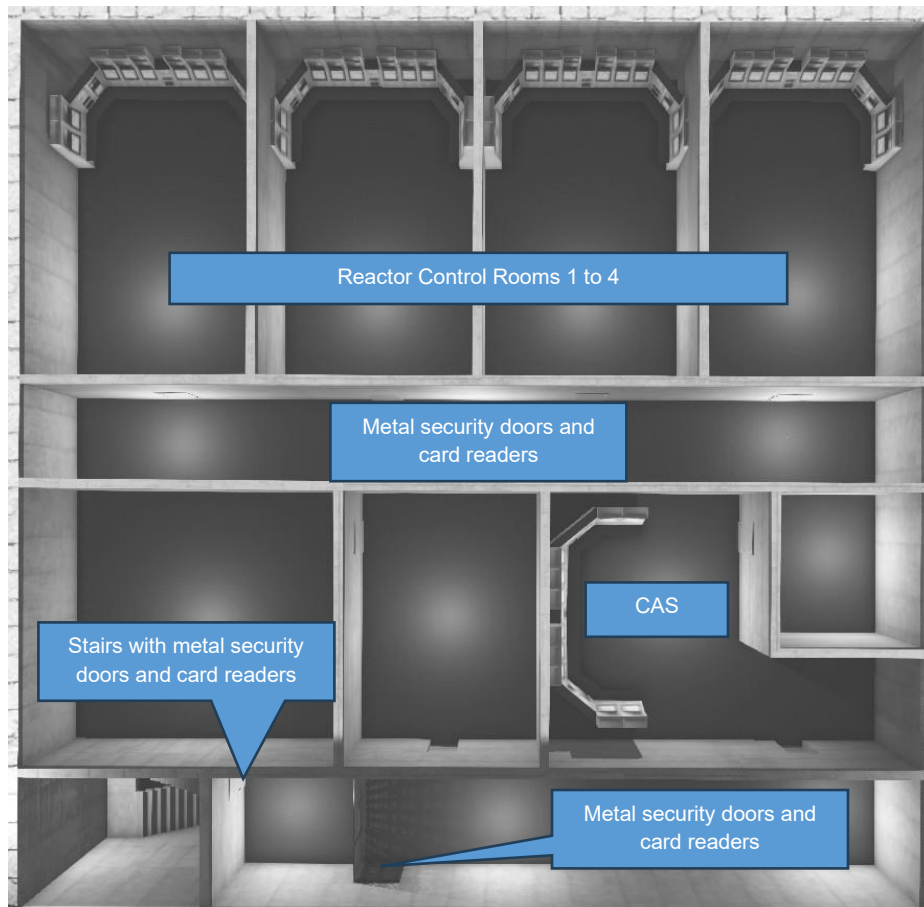


Figure 15. Physical security elements in office building, second floor

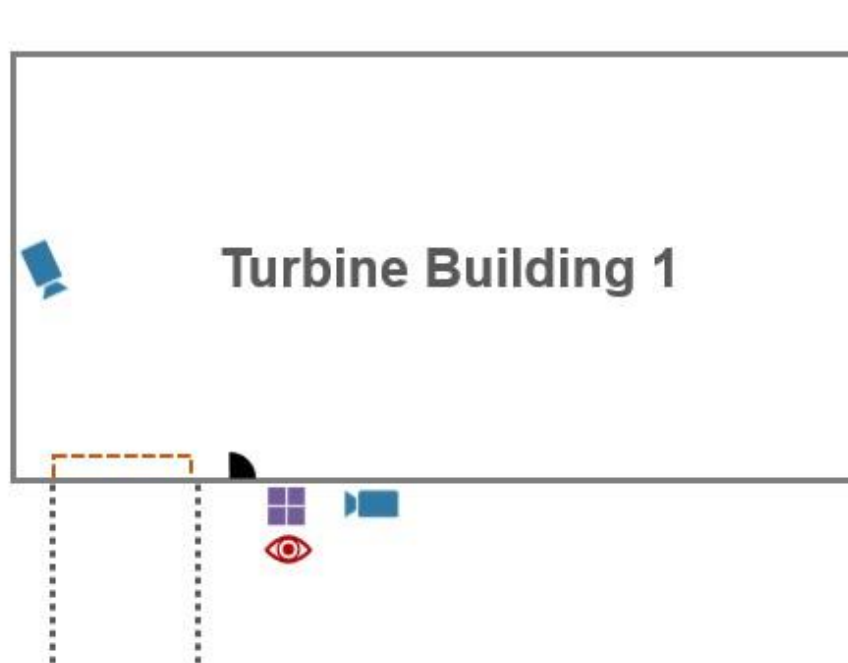


Figure 16. Physical security elements in turbine building 1, ground floor

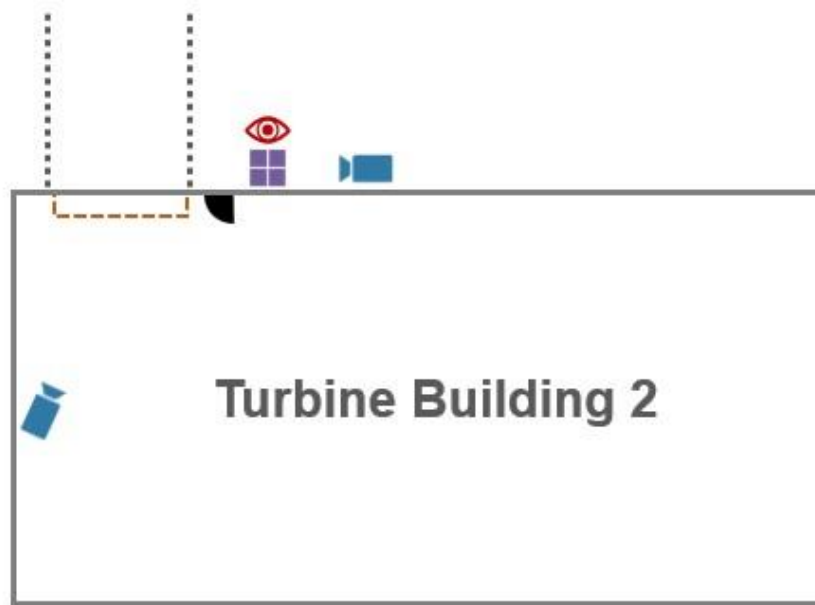


Figure 17. Physical security elements in turbine building 2, ground floor

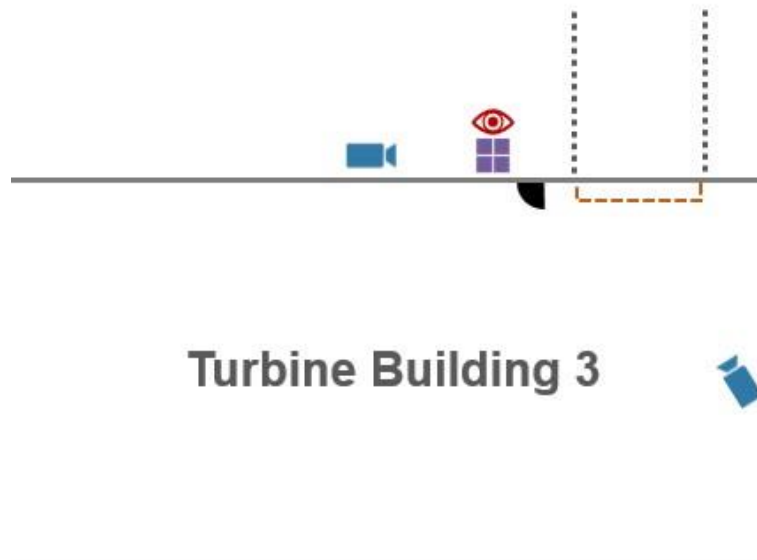


Figure 18. Physical security elements in turbine building 3, ground floor

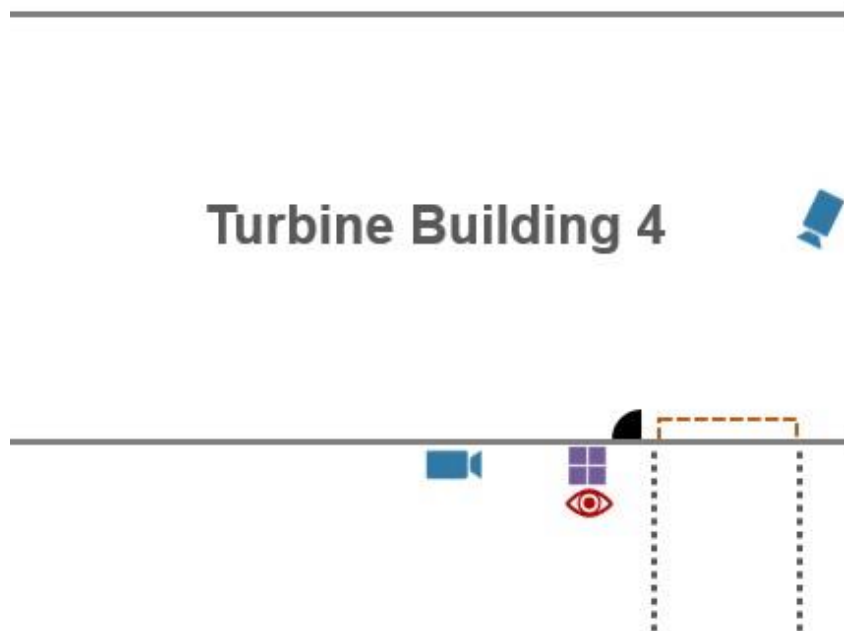


Figure 19. Physical security elements in turbine building 4, ground floor

5 PATHWAY ANALYSIS

Pathway analysis is used to evaluate the relative balance of security system components in a PPS. Probabilities of sensing (P_S), assessment (P_A), and delay are characterized and used in a systematic model to determine the pathways that offer adversaries the least amount of detection and delay. The combined multiplicative value of probability of sensing and probability of assessment results in probability of detection (P_D).⁷

The critical detection point (CDP) is the last point with a non-zero P_D , on an adversary path where the minimum delay from remaining tasks on the path is greater than or equal to the RF time. The location of the CDP can be used by the analyst to measure the balance between delay, detection, and response to measure balance in the system and to determine the pathway with the lowest probability of interruption (P_I) for the system. P_I is defined as the cumulative P_D up to and including the CDP.

As an example, using the formula listed above, if there are three detection points (P_{D1} , P_{D2} , and P_{D3}) along an adversary path where the CDP is located at layer 3, P_I is calculated as follows:

$$P_I = P_{D1} + ((1 - P_{D1}) * P_{D2}) + ((1 - P_{D1}) * (1 - P_{D2}) * P_{D3})$$

The critical adversary pathway analysis elements listed above equate to the eventual adversary pathways selected and eventually used in scenario development.

Additionally, for the purpose of this vulnerability assessment report, pathways were analyzed with PathTrace software but written/selected in a balanced approach with respect to cardinal direction. To fully analyze adversary scenarios, pathways were chosen evenly among the four main cardinal directions leading to the site. This purpose was to attack the site from different directions to analyze various detection/response capabilities.

Consideration was given to subject matter expert (SME) opinions on creditable adversary pathways given the DBT. PathTrace has the ability to compute four adversary pathway types:

1. Draw Path: Tool to hand-draw adversary pathways.
2. Quickest Path: Tool that generates the lowest total delay pathway between the offsite and the user's cursor.

⁷ In Department of Energy terminology, P_S for an adversary traversal or act is the probability that a given sensor alarms due to that presence or action of the adversary. Separately, P_A is the probability that, given the sensing occurs, the adversary will be assessed and communicated to the RF to initiate a response. For the purposes of this report, P_S will not be used as NRC uses P_D in place of this term. The NRC's characterization of P_D does not include assessment.

3. Stealthiest Path: Tool that generates the lowest total P_D pathway between the offsite and the user's cursor.
4. Most Vulnerable Path: Tool that generates the lowest total P_I pathway between the offsite and the user's cursor.

The fastest and most vulnerable path types were generated and recorded. Then SME input was considered if these pathways were deemed not creditable for the given DBT.

PathTrace also generates the following calculations for each adversary pathway computed:

- Total P_D : Probability that, at any point along the pathway, the adversary is both detected and assessed.
- Total P_I : Probability that the adversary will be detected and assessed in time for responders to interrupt the adversary.
- Total Delay: Sum of all defeat and traversal times of barriers, detection areas, jumps, and targets along the pathway.

The above data was recorded for each adversary pathway and considered in the final selection of the scenarios that were run for this report.

5.1 Probability of Interruption

P_I is a figure of merit (FoM) that measures the ability of the responders to arrive in time to interrupt the adversary before they complete their objective. In addition to the adversaries' ability to defeat elements of the PPS, as captured by P_D and the delay time, P_I depends on the officers' RF time. For the purposes of this report, interruption occurs when two armed officers arrive at a location to interrupt the adversaries. Based on performance metrics, the maximum time for two armed officers to respond anywhere onsite is 60 seconds.

5.2 Minimized Delay Pathway

A minimized delay pathway determination (fastest path in PathTrace) is a process to evaluate the balance of delay systems at the site. This process is not used to calculate P_I but is used to understand the FoM provided by barriers in the system. It may also be used to assist the analyst in scenario development. This methodology does not consider detection as part of the pathway determination. This pathway is determined as follows:

1. Determine the overall adversary task time.
2. Determine protective force response strategy.
3. From the starting point of the task, the adversary will make choices to minimize delay, regardless of the likelihood of detection by the protection system.
4. The total time for the pathway is the cumulative delay after the initial detection.

6 SCENARIO DEVELOPMENT

To create a representative suite of scenarios reflecting a range of adversary capabilities outlined in the DBT, a multidisciplinary scenario development team (SDT) was formed. This team included SMEs in reactor operations, PPSs, insider threats, site protection strategies, explosive effects, modeling and simulation software, and adversary tactics. The SDT developed scenarios to accurately evaluate the effectiveness of the security protection system at the SMRF.

The SDT coordinated to analyze adversary pathways and sabotage targets, leveraging SME expertise and vital area information in the hypothetical facility. Recognizing that the DBT considers both active and passive insiders, the scenarios were crafted with an understanding of the facility layout and RF locations and timelines, accommodating the potential involvement of a passive insider. Scenario 4 features an active, non-violent insider who disables doors and obstructs response efforts, to capture the involvement of a potential active insider. In total, four scenarios were developed to assess various pathways, adversary capabilities, protection systems, and RF elements.

6.1 Scenario 1—Special Nuclear Material Vault

This scenario is of an outsider attack on the special nuclear material vault. The adversary mission is to infiltrate the plant and commit theft of special nuclear material. This is an overt attack using force to steal the special nuclear material. As such, the scenario does not conclude until adversaries have left the site with the material.

6.2 Scenario 2—Main Control Rooms

This scenario is of an attack on the main control room. The adversary mission for this scenario is to enter the main control room and effect sabotage of the reactor via the systems in the control room. This is an overt attack that uses force to capture the main control room for sabotage. As such, the scenario concludes when the adversaries have sabotaged the reactor; adversary success is not dependent on them leaving the facility.

6.3 Scenario 3—Auxiliary Feedwater Pumps

This scenario is of an attack on the auxiliary feedwater pumps. The adversary mission is to sabotage a reactor via destruction of its auxiliary feedwater pumps. This is an overt attack that uses force to gain access to, and destroy, the auxiliary feedwater pumps for one reactor. This scenario concludes when adversaries destroy both targeted feedwater pumps.

6.4 Scenario 4—Auxiliary Feedwater Pumps with Active Insider

This scenario is of an attack on the auxiliary feedwater pumps with the support of an active insider. The adversary mission is to sabotage a reactor via destruction of its auxiliary feedwater pumps. This is a mixed covert and overt attack that uses both stealth and force to gain access to, and destroy, the auxiliary feedwater pumps for one reactor. The active insider assists by

manipulating security measures to make it more challenging for responders to defend the targeted pumps.

7 NEUTRALIZATION METHODOLOGY

The probability of neutralization (P_N) is a numeric value representing the probability the RF can kill or render an adversary incapable of completing their attack objective(s), following interruption. P_N is based on RF success criteria derived from a combination of the results from computer simulations.

Computer simulations were conducted in Dante⁸, a modeling and simulation tool that generates statistical data for force-on-force exercises. Dante simulates adversary attacks on sites of concern using Monte Carlo techniques. Dante assists in computing P_N by processing large simulation outcomes through batch-mode executions. P_N results are supported by performance testing results and other assurance activities conducted, particularly assurance activities related to RF duties and response plans. The cumulative data gathered is analyzed to ensure key assumptions used in computer models and the resultant P_N value are supported

In Dante, the terrain for the site is developed as a high-fidelity 3D model with key features such as roads, fences, buildings, trees, and other terrain elements. On top of the terrain, the analyst configures sensor systems such as cameras, fence sensors, and radars so the elements of detection and assessment are modeled. The analyst then adds the blue RF to include response teams in the field and command elements such as the central alarm station and the blue commander to direct the response

Scenarios 1–4 were programmed into the Dante tool; and, after executing 250 batch runs for each scenario, wins and losses were collected. P_N for each of the scenarios was calculated as wins/tests.

⁸ Sandia National Laboratories has additional information on Dante at <https://dante.sandia.gov/>

8 SYSTEM EFFECTIVENESS METHODOLOGY

Following the completion of the scenario development and P_N analyses, the site will conduct the analyses of and calculate a probability of effectiveness (P_E) value for each scenario to determine the PPS effectiveness. This section describes how to calculate P_E when separate P_I and P_N have been calculated as described previously.

The method to calculate P_E is to identify the P_I and an appropriate P_N . P_E is determined by the following formula:

$$P_E = P_I * P_N$$

Utilizing modeling and simulation software provides the opportunity to conduct as many iterations as desired to evaluate probabilities. The software then provides a number of wins and losses. These results may then evaluate P_E' , which includes confidence considerations, as desired. This may be accomplished with an appropriate confidence level integrated using a formula such as:

$$P_E' = \frac{\text{Wins}}{\text{Tests}} * (1 - \frac{1}{2\sqrt{\text{Tests}}})$$

This process evaluates a value of P_E' for each of the analyzed scenarios. Neither P_E nor P_E' should be averaged; the values should be individually reported. The lowest value for P_E' corresponds to the most vulnerable scenario for the facility.

9 CONCLUSION

This facility demonstrates a methodology for determining the effectiveness of a facility with a high degree of security. This level of security is primarily attributed to the facility's design, which integrates security measures during the design phase rather than retrofitting them post construction. Key findings from the analysis include:

1. **Facility Design and Security Integration:** The facility design incorporates robust security features, such as below-grade reactor placement, reinforced concrete structures, and comprehensive access control measures, which enhance the facility's security posture.
2. **Environmental and Operational Conditions:** The facility's location in a semiarid region with minimal vegetation and wildlife, coupled with its operational schedule, supports a controlled and consistent environment for security operations.
3. **Reactor and Safety Systems:** The reactor design, featuring natural circulation cooling, ECCS, and dry cooling radiators, supports the safe and efficient operation of the reactors. The integration of multiple safety-related systems, such as the SCIS and the standby liquid control system, provides robust resilience to potential threats.
4. **Security Measures and RF:** The facility employs a layered security approach, including PIDAS, redundant communication systems, and a well-trained onsite response team. The RF's equipment, training, and deployment strategies are designed to effectively counter various adversary scenarios.
5. **Scenario Analysis and P_N:** The facility's security system was rigorously tested through multiple adversary scenarios, including theft, sabotage, and insider threats.